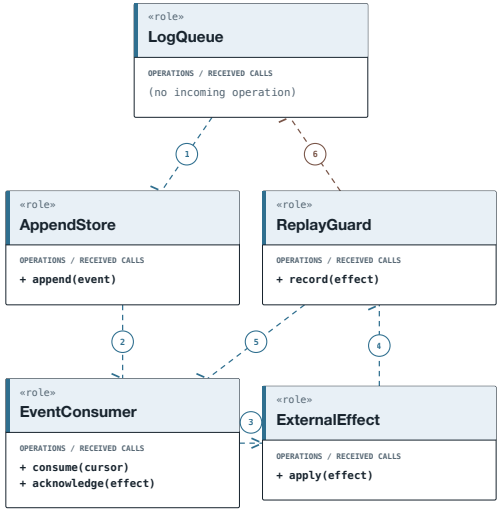
# P13 Log-as-queue loops

**Primitive / UML CLASS VIEW / ANALYTICAL MODEL**

Drive work from a durable execution record.

**UML CLASS VIEW**

dashed arrow: call dependency · brown: return/failure · analytical, not an observed trace




**POINTCUT / PATTERN INTERFACE**

| | |
|---|---|
| CONTEXT | New work is represented by durable append-only records. |
| TRIGGER | A consumer polls or subscribes to unseen records. |
| STRUCTURE | Event identity, append order, consumer cursor, and acknowledgement. |
| EFFECT | Work advances from the log without replaying already completed effects. |

**CONTRACT / REQUIRED BEHAVIOR**

| | |
|---|---|
| REQUIRES | Records have stable identity and consumers retain progress. |
| ENSURES | Each effectful unit has an idempotency or acknowledgement rule. |
| INVARIANT | Consumer crash does not erase unacknowledged work. |
| VIOLATION | Duplicate delivery is deduplicated or documented as at-least-once. |

**RELATIONSHIPS / ANALYTICAL CALLS**

number · caller → callee · message

1 LogQueue → AppendStore append(event)
2 AppendStore → EventConsumer consume(cursor)
3 EventConsumer → ExternalEffect apply(effect)
4 ExternalEffect → ReplayGuard record(effect)
5 ReplayGuard → EventConsumer acknowledge(effect)
6 ReplayGuard ↩ LogQueue replay(cursor)

**VARIATION / TEST / NEGATIVE CONTROL**

| | |
|---|---|
| VARIATION | SQL log, file log, queue adapter, polling, and projection. |
| TEST_OBLIGATION | Test crash before ack, duplicate delivery, out-of-order reads, and replay. |
| NEGATIVE_CONTROL | An in-memory queue is not log-as-queue. |